

# Class 09: Candy Mini Project

Barry (PID: 911)

## Table of contents

|                                               |    |
|-----------------------------------------------|----|
| Background . . . . .                          | 1  |
| Data import . . . . .                         | 1  |
| Exploratory analysis . . . . .                | 4  |
| Overall Candy Rankings . . . . .              | 6  |
| Taking a look at pricepercent . . . . .       | 16 |
| Exploring the correlation structure . . . . . | 17 |
| Principal Component Analysis . . . . .        | 18 |

## Background

Today we are taking a wee detour to analyze a fun data-set (that we have more intrinsic insight into) with the most useful analysis method we have learned thus far - Principal Component Analysis (PCA)

## Data import

The data is all related to Halloween candy and is from the 538 website.

```
candy <- read.csv("candy-data.txt", row.names = 1)
head(candy)
```

|              | chocolate | fruity | caramel | peanutyalmondy | nougat | crispedricewafer |
|--------------|-----------|--------|---------|----------------|--------|------------------|
| 100 Grand    | 1         | 0      | 1       | 0              | 0      | 1                |
| 3 Musketeers | 1         | 0      | 0       | 0              | 1      | 0                |
| One dime     | 0         | 0      | 0       | 0              | 0      | 0                |
| One quarter  | 0         | 0      | 0       | 0              | 0      | 0                |
| Air Heads    | 0         | 1      | 0       | 0              | 0      | 0                |

|              |      |     |          |              |              |            |
|--------------|------|-----|----------|--------------|--------------|------------|
| Almond Joy   | 1    | 0   | 0        | 1            | 0            | 0          |
|              | hard | bar | pluribus | sugarpercent | pricepercent | winpercent |
| 100 Grand    | 0    | 1   | 0        | 0.732        | 0.860        | 66.97173   |
| 3 Musketeers | 0    | 1   | 0        | 0.604        | 0.511        | 67.60294   |
| One dime     | 0    | 0   | 0        | 0.011        | 0.116        | 32.26109   |
| One quarter  | 0    | 0   | 0        | 0.011        | 0.511        | 46.11650   |
| Air Heads    | 0    | 0   | 0        | 0.906        | 0.511        | 52.34146   |
| Almond Joy   | 0    | 1   | 0        | 0.465        | 0.767        | 50.34755   |

Q1. How many different candy types are in this dataset?

Q2. How many fruity candy types are in the dataset?

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
library(dplyr)
```

Warning: package 'dplyr' was built under R version 4.4.3

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy |>
  filter(row.names(candy)=="Air Heads") |>
  select(pricepercent, sugarpercent, winpercent)
```

|           |              |              |            |
|-----------|--------------|--------------|------------|
|           | pricepercent | sugarpercent | winpercent |
| Air Heads | 0.511        | 0.906        | 52.34146   |

Q4. What is the winpercent value for “Kit Kat”?

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

There is a useful `skim()` function in the **skimr** package that can help give you a quick overview of a given dataset. Let's install this package and try it on our candy data.

```
skimr::skim(candy)
```

Table 1: Data summary

|                                   |       |
|-----------------------------------|-------|
| Name                              | candy |
| Number of rows                    | 85    |
| Number of columns                 | 12    |
| Column type frequency:<br>numeric | 12    |
| Group variables                   | None  |

#### Variable type: numeric

| skim_vari-<br>able    | n_miss-<br>ing | com-<br>plete_rate | mean  | sd    | p0    | p25   | p50   | p75   | p100  | hist |
|-----------------------|----------------|--------------------|-------|-------|-------|-------|-------|-------|-------|------|
| chocolate             | 0              | 1                  | 0.44  | 0.50  | 0.00  | 0.00  | 0.00  | 1.00  | 1.00  |      |
| fruity                | 0              | 1                  | 0.45  | 0.50  | 0.00  | 0.00  | 0.00  | 1.00  | 1.00  |      |
| caramel               | 0              | 1                  | 0.16  | 0.37  | 0.00  | 0.00  | 0.00  | 0.00  | 1.00  |      |
| peanutyal-<br>mondy   | 0              | 1                  | 0.16  | 0.37  | 0.00  | 0.00  | 0.00  | 0.00  | 1.00  |      |
| nougat                | 0              | 1                  | 0.08  | 0.28  | 0.00  | 0.00  | 0.00  | 0.00  | 1.00  |      |
| crispedrice-<br>wafer | 0              | 1                  | 0.08  | 0.28  | 0.00  | 0.00  | 0.00  | 0.00  | 1.00  |      |
| hard                  | 0              | 1                  | 0.18  | 0.38  | 0.00  | 0.00  | 0.00  | 0.00  | 1.00  |      |
| bar                   | 0              | 1                  | 0.25  | 0.43  | 0.00  | 0.00  | 0.00  | 0.00  | 1.00  |      |
| pluribus              | 0              | 1                  | 0.52  | 0.50  | 0.00  | 0.00  | 1.00  | 1.00  | 1.00  |      |
| sugarpercent          | 0              | 1                  | 0.48  | 0.28  | 0.01  | 0.22  | 0.47  | 0.73  | 0.99  |      |
| pricepercent          | 0              | 1                  | 0.47  | 0.29  | 0.01  | 0.26  | 0.47  | 0.65  | 0.98  |      |
| winpercent            | 0              | 1                  | 50.32 | 14.71 | 22.45 | 39.14 | 47.83 | 59.86 | 84.18 |      |

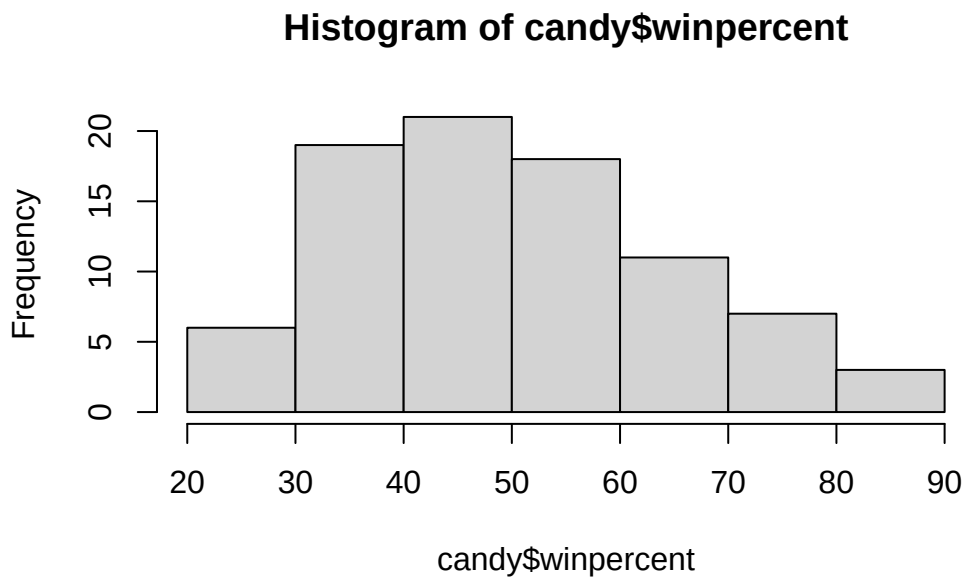
Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Q7. What do you think a zero and one represent for the `candy$chocolate` column?

## Exploratory analysis

Q8. Plot a histogram of `winpercent` values using both base R and ggplot2.

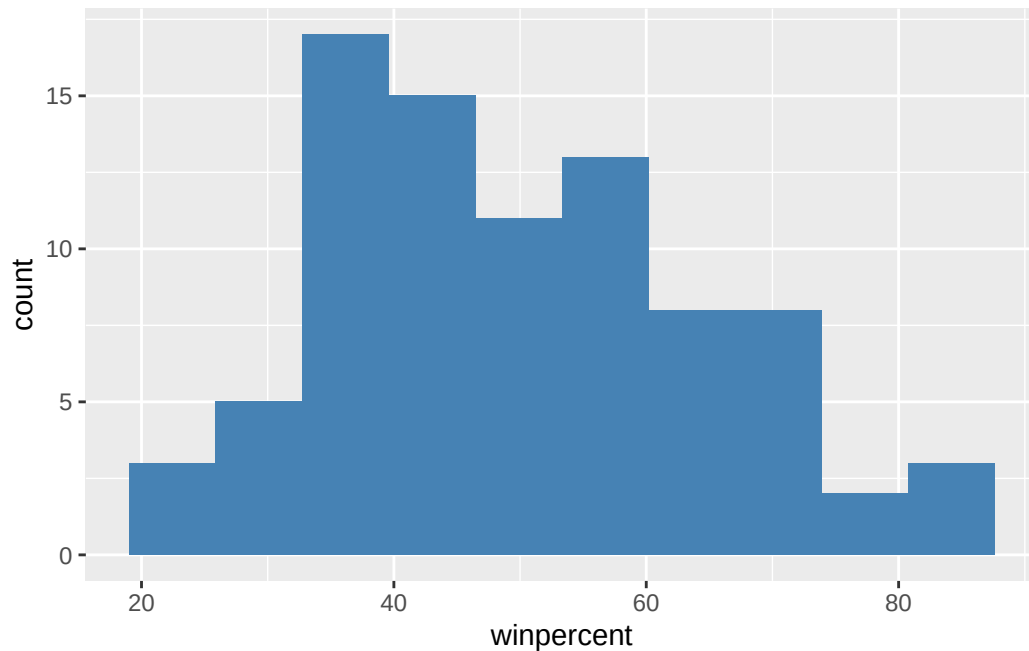
```
hist(candy$winpercent)
```



```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.4.3

```
ggplot(candy) +  
  aes(winpercent) +  
  geom_histogram(bins=10, fill="steelblue")
```



Q9. Is the distribution of winpercent values symmetrical?

No

Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

```
summary(candy$winpercent)
```

| Min.  | 1st Qu. | Median | Mean  | 3rd Qu. | Max.  |
|-------|---------|--------|-------|---------|-------|
| 22.45 | 39.14   | 47.83  | 50.32 | 59.86   | 84.18 |

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
choc.ind <- as.logical(candy$chocolate)
choc.candy <- candy[choc.ind, ]
choc.win <- choc.candy$winpercent
mean(choc.win)
```

```
[1] 60.92153
```

```
fruit.ind <- as.logical(candy$fruity)
fruit.candy <- candy[fruit.ind,]
fruit.win <- fruit.candy$winpercent
mean(fruit.win)
```

```
[1] 44.11974
```

Q12. Is this difference statistically significant?

```
t.test(choc.win, fruit.win)
```

Welch Two Sample t-test

```
data:  choc.win and fruit.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

## Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```
candy |>
  arrange(winpercent) |>
  select(winpercent) |>
  head(5)
```

|                    | winpercent |
|--------------------|------------|
| Nik L Nip          | 22.44534   |
| Boston Baked Beans | 23.41782   |
| Chiclets           | 24.52499   |
| Super Bubble       | 27.30386   |
| Jawbusters         | 28.12744   |

```
x <- c(5,10,1)
sort(x)
```

```
[1] 1 5 10
```

```
x <- c(5,10,1)
order(x)
```

```
[1] 3 1 2
```

```
x[ order(x) ]
```

```
[1] 1 5 10
```

```
head( candy[ order(candy$winpercent), ] )
```

|                    | chocolate | fruity | caramel | peanut | almond | nougat |
|--------------------|-----------|--------|---------|--------|--------|--------|
| Nik L Nip          | 0         | 1      | 0       |        | 0      | 0      |
| Boston Baked Beans | 0         | 0      | 0       |        | 1      | 0      |
| Chiclets           | 0         | 1      | 0       |        | 0      | 0      |
| Super Bubble       | 0         | 1      | 0       |        | 0      | 0      |
| Jawbusters         | 0         | 1      | 0       |        | 0      | 0      |
| Root Beer Barrels  | 0         | 0      | 0       |        | 0      | 0      |

|                    | crisped | rice | wafer | hard | bar | pluribus | sugar | percent | price | percent |
|--------------------|---------|------|-------|------|-----|----------|-------|---------|-------|---------|
| Nik L Nip          |         | 0    | 0     | 0    |     | 1        |       | 0.197   |       | 0.976   |
| Boston Baked Beans |         | 0    | 0     | 0    |     | 1        |       | 0.313   |       | 0.511   |
| Chiclets           |         | 0    | 0     | 0    |     | 1        |       | 0.046   |       | 0.325   |
| Super Bubble       |         | 0    | 0     | 0    |     | 0        |       | 0.162   |       | 0.116   |
| Jawbusters         |         | 0    | 1     | 0    |     | 1        |       | 0.093   |       | 0.511   |
| Root Beer Barrels  |         | 0    | 1     | 0    |     | 1        |       | 0.732   |       | 0.069   |

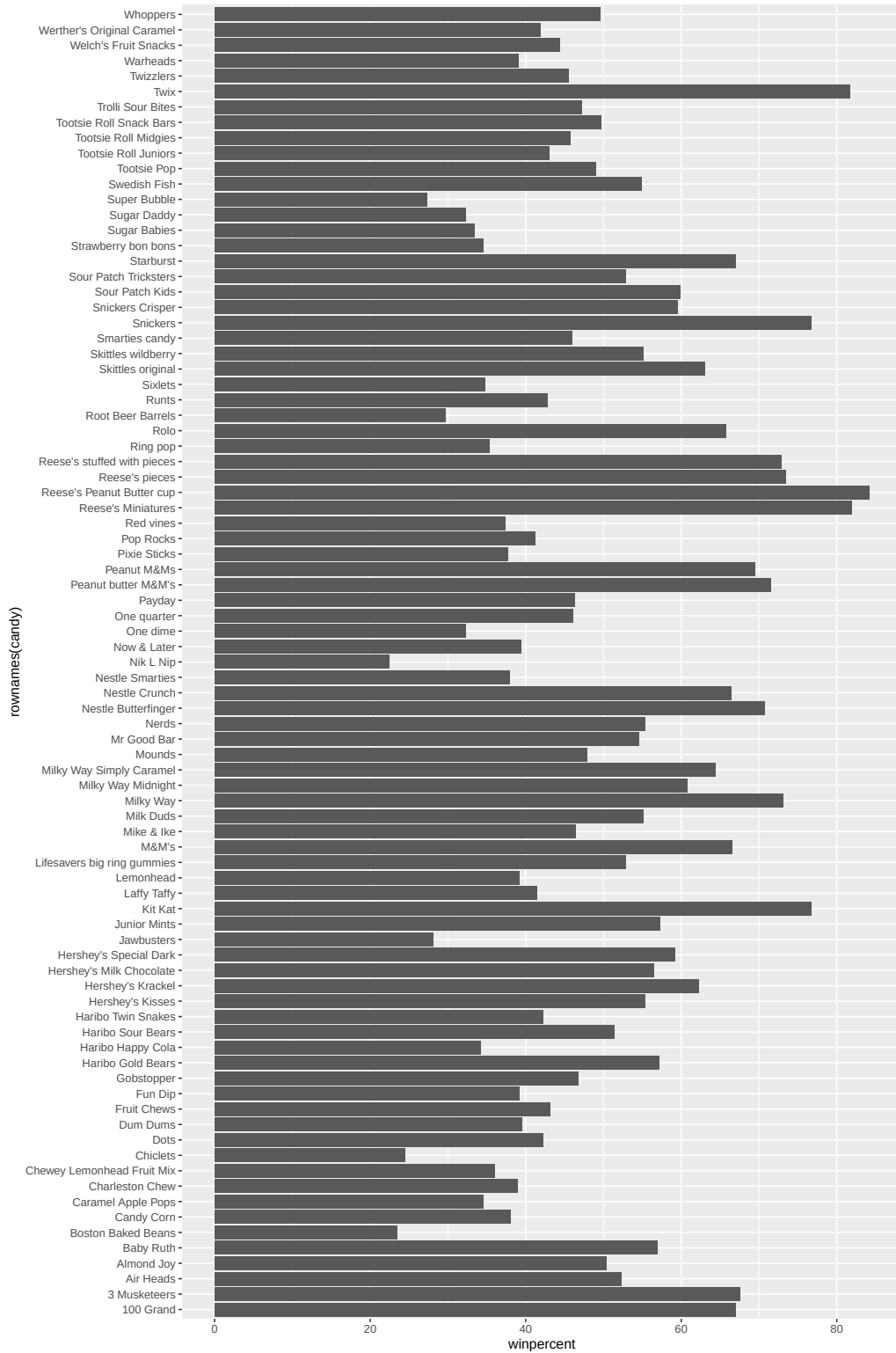
|                    | winpercent |
|--------------------|------------|
| Nik L Nip          | 22.44534   |
| Boston Baked Beans | 23.41782   |
| Chiclets           | 24.52499   |
| Super Bubble       | 27.30386   |
| Jawbusters         | 28.12744   |
| Root Beer Barrels  | 29.70369   |

Q14. What are the top 5 all time favorite candy types out of this set?

Q15. Make a first barplot of candy ranking based on winpercent values.

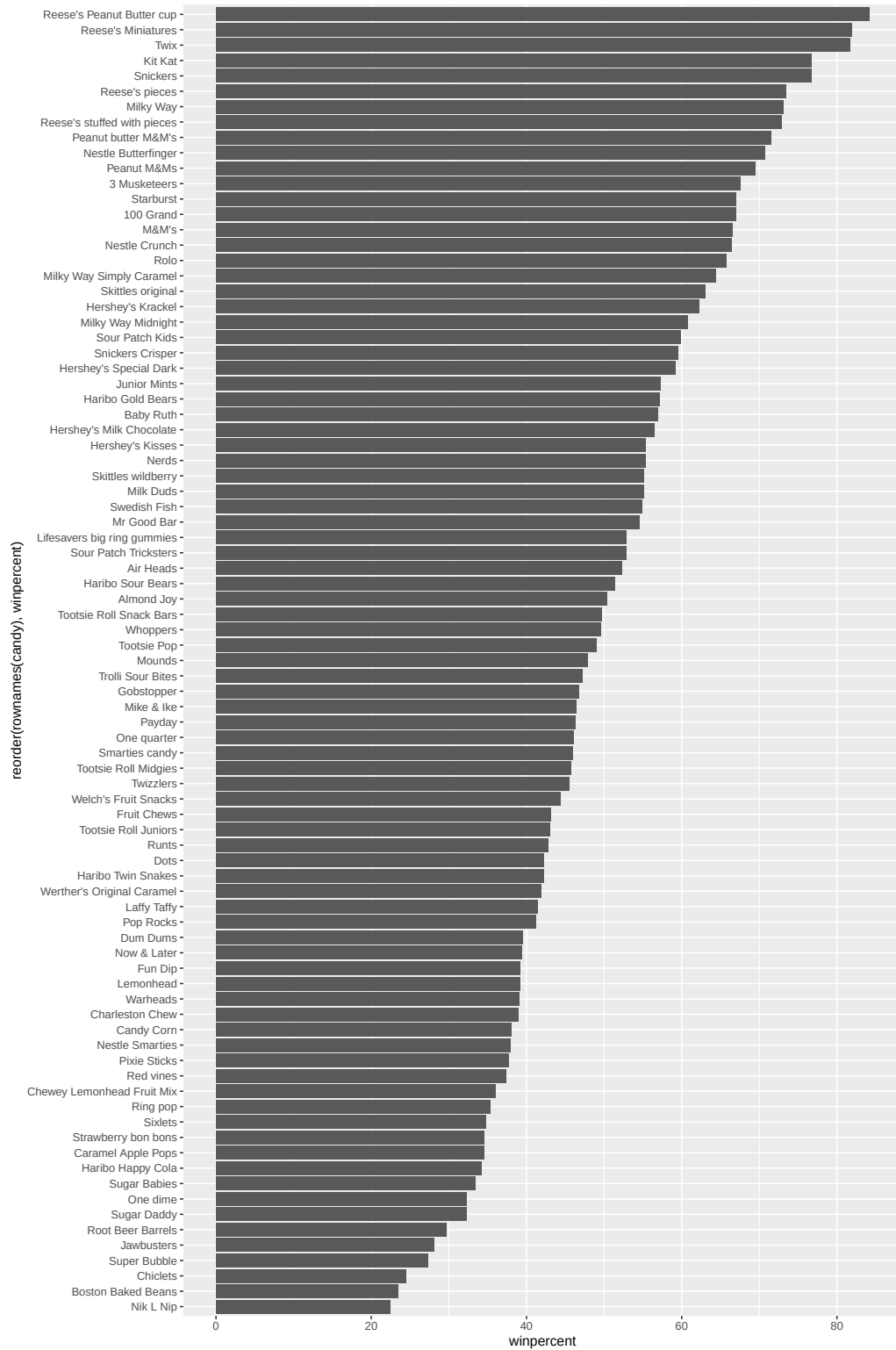
```
ggplot(candy) +  
  aes(winpercent, rownames(candy)) +  
  geom_col()
```





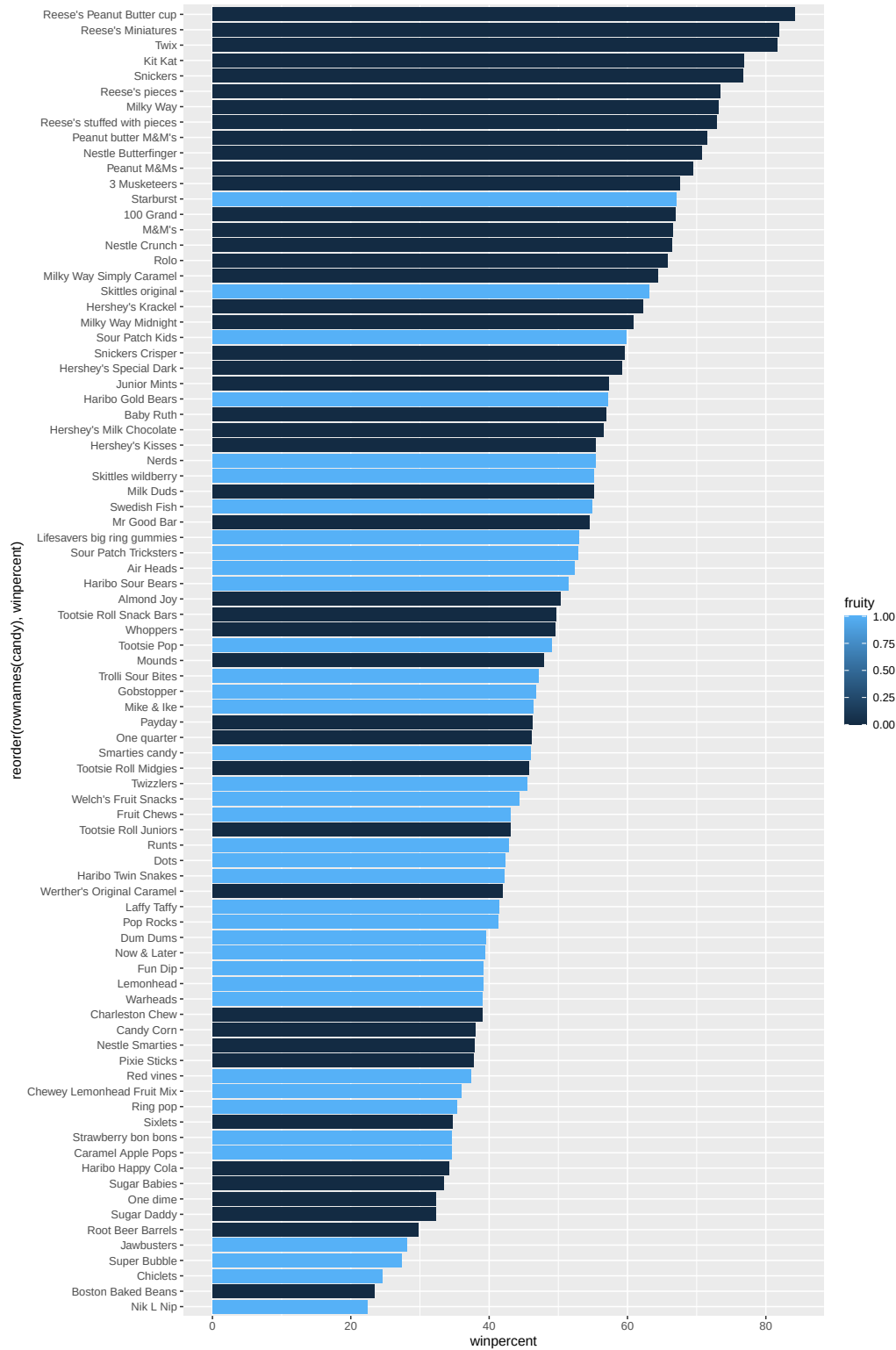
Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

```
ggplot(candy) +  
  aes(winpercent, reorder(rownames(candy), winpercent)) +  
  geom_col()
```



Adding some color

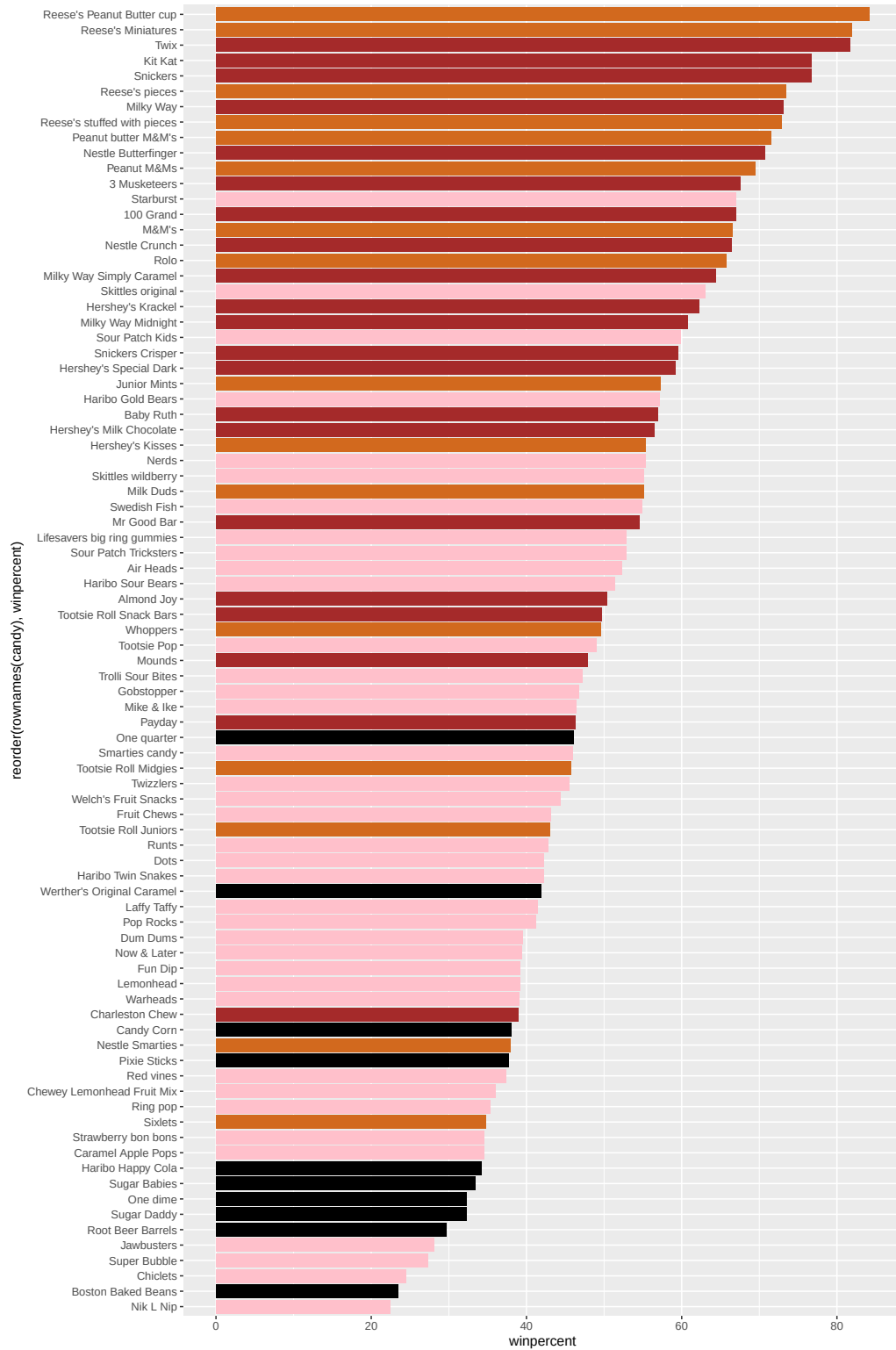
```
ggplot(candy) +  
  aes(winpercent,  
       reorder(rownames(candy), winpercent),  
       fill=fruity) +  
  geom_col()
```



```
mycols <- rep("black", nrow(candy))

## chocolate candy in chocolate color
mycols[ as.logical(candy$chocolate) ] <- "chocolate"
# candy bars in brown
mycols[ as.logical(candy$bar) ] <- "brown"
# fruity candy in pink
mycols[ as.logical(candy$fruity) ] <- "pink"

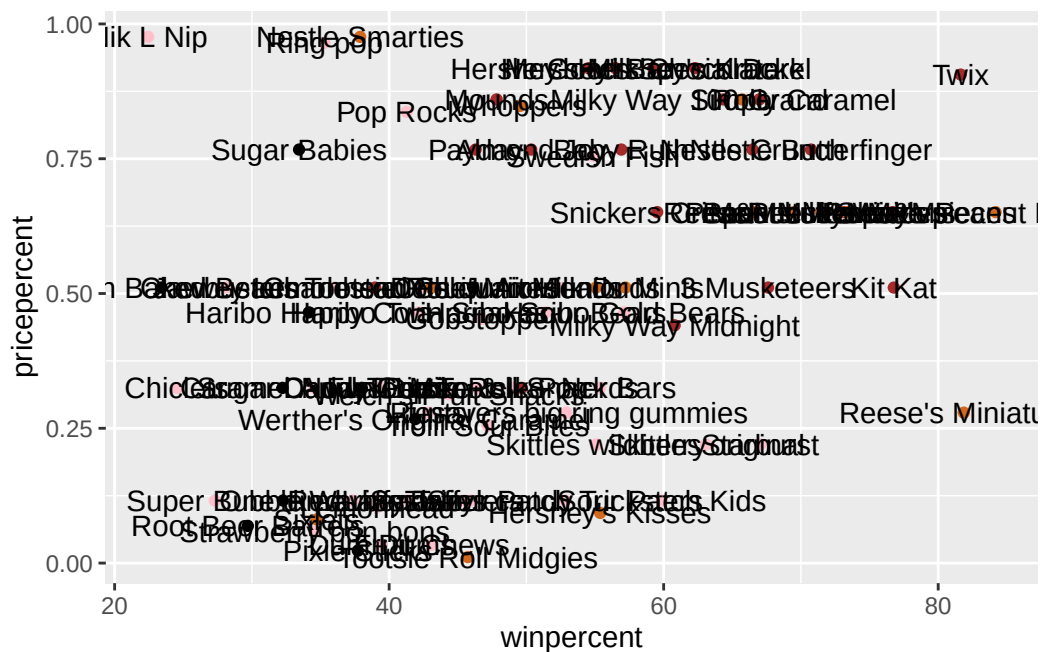
ggplot(candy) +
  aes(winpercent,
      reorder(rownames(candy), winpercent)) +
  geom_col(fill=mycols)
```



## Taking a look at pricepercent

Make a plot of winpercent vs pricepercent

```
ggplot(candy) +
  aes(winpercent,
       pricepercent,
       label=rownames(candy)) +
  geom_point(col=mycols) +
  geom_text()
```



We can fix the label text overplotting with an add-on package called **ggrepel** and it's `geom_text_repel()`

```
# fruity candy in pink
mycols[ as.logical(candy$fruity) ] <- "red" #"pink"
```

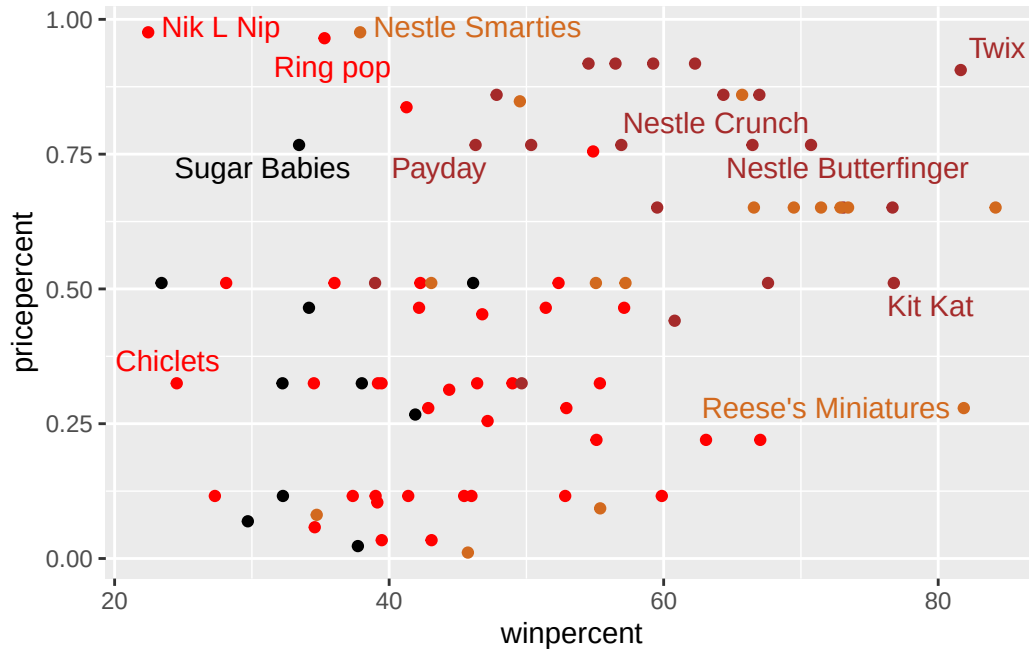
```
library(ggrepel)

ggplot(candy) +
  aes(winpercent,
       pricepercent,
       label=rownames(candy)) +
```



```
geom_point(col=mycols) +  
geom_text_repel(col=mycols, max.overlaps = 5)
```

Warning: ggrepel: 74 unlabeled data points (too many overlaps). Consider increasing max.overlaps



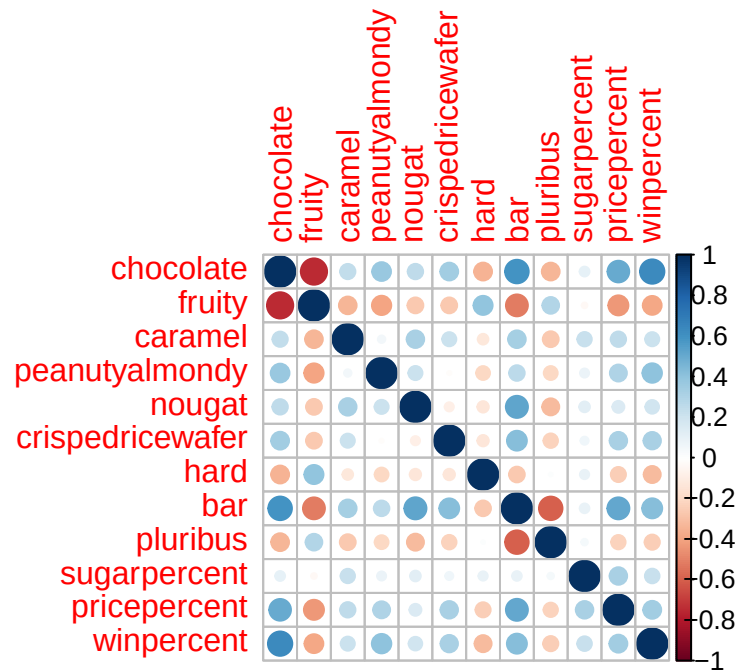
## Exploring the correlation structure

We can calculate the pair-wise correlation of all our columns

```
cij <- cor(candy)  
library(corrplot)
```

corrplot 0.95 loaded

```
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Q23. Use your corplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

## Principal Component Analysis

In this case we want to be sure to set `scale=TRUE` argument for `prcomp()` because we have one variable, `winpercent`, that is on a very different scale than all others and would otherwise dominate our PCA results.

```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

Importance of components:

|                        | PC1    | PC2    | PC3    | PC4     | PC5    | PC6     | PC7     |
|------------------------|--------|--------|--------|---------|--------|---------|---------|
| Standard deviation     | 2.0788 | 1.1378 | 1.1092 | 1.07533 | 0.9518 | 0.81923 | 0.81530 |
| Proportion of Variance | 0.3601 | 0.1079 | 0.1025 | 0.09636 | 0.0755 | 0.05593 | 0.05539 |
| Cumulative Proportion  | 0.3601 | 0.4680 | 0.5705 | 0.66688 | 0.7424 | 0.79830 | 0.85369 |

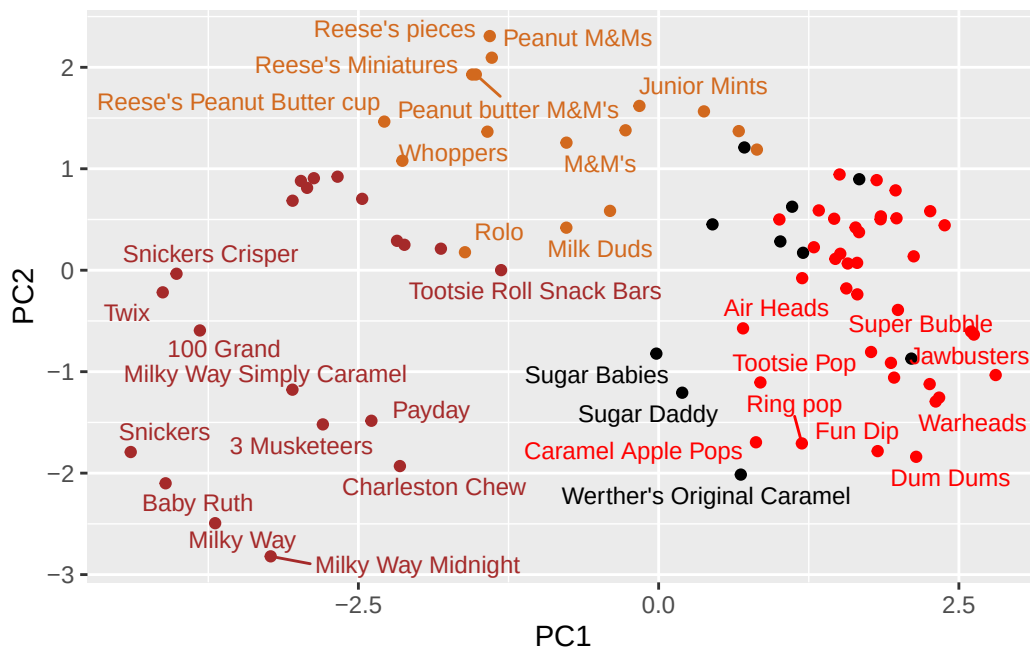
|  | PC8 | PC9 | PC10 | PC11 | PC12 |
|--|-----|-----|------|------|------|
|--|-----|-----|------|------|------|

|                        |         |         |         |         |         |
|------------------------|---------|---------|---------|---------|---------|
| Standard deviation     | 0.74530 | 0.67824 | 0.62349 | 0.43974 | 0.39760 |
| Proportion of Variance | 0.04629 | 0.03833 | 0.03239 | 0.01611 | 0.01317 |
| Cumulative Proportion  | 0.89998 | 0.93832 | 0.97071 | 0.98683 | 1.00000 |

First major result figure is the “score plot” of PC1 vs PC2 - how the different candy are related to each other on our new PC axis:

```
ggplot(pca$x) +
  aes(PC1, PC2, label=row.names(pca$x)) +
  geom_point(col=mycols) +
  geom_text_repel(size=3.3, col=mycols, max.overlaps = 7)
```

Warning: ggrepel: 51 unlabeled data points (too many overlaps). Consider increasing max.overlaps



The second major results figure from PCA is the so-called “loadings plot”

```
ggplot(pca$rotation) +
  aes(PC1,
    reorder( row.names(pca$rotation), PC1)) +
  geom_col()
```

